DataScienceProject

Izzy Mika

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Side Note: I still have additional tests (chi square goodness of fit test / other graphs if those are needed or useful.)

Set-up

```
library(readxl)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Datasets

```
filipino_stats <- read_xlsx("C:/Users/pikmi/Documents/MetroState/25 Data 211 (Data Science)/Data Projec
```

Data clean-up

```
# Clean up data
filipino_stats <- filipino_stats %>%
  filter(Keep==1) %>%                    #Keep tracked words (ignore eg function words, proper nouns)
  group_by(Day) %>%                      #Group by day / get numbers per day
  summarize(
    day_count = n()                      #Total words per day
    ,Y_count = sum(Correct_YN == 'Y')    #Total correct answers per day
```

```
    ,N_count = sum(Correct_YN == 'N')      #Total incorrect answers per day
    ,percent_Y = (sum(Correct_YN == 'Y')/day_count)    #Percent correct
    ,percent_N = (sum(Correct_YN == 'N')/day_count)    #Percent incorrect

  )

# Data - Removing Day 8, 9, 10
#---------------------------------
# Didn't realize this until later - the video used for part of 8/10
#   (and all of 9) interviews people mostly speaking Bisaya(/Binisaya), which is
#   a related but diff language that I haven't studied before/was not studying
#   via the course I had. Removing from final data set to avoid wonky results.

tagalog_stats <- filipino_stats %>%
  filter(Day < 8 | Day > 10)


#tagalog_stats
#tagalog_stats$Y_count
#tagalog_stats$percent_Y
```

Prop Test - Comparing Correct vs Incorrect Answers

I want to see if studying Tagalog via the LearningTagalog course for 20 min every day (over 14 days) increases the percentage of words I understand over time when reading/watching conversational Tagalog.

I'm using prop testing to test whether the proportion of correct answers is significantly greater than proportion of wrong answers.

P_Y - Proportion of correct answers ("Y")

P_N - Proportion of incorrect answers ("N")

$H_0 : P_Y = P_N$

$H_a : P_Y > P_N$

```
# Test - is proportion of correct answers greater than proportion of wrong answers?
# Sum all Ys, sum all Ns, see if Y > N

allY <- sum(tagalog_stats$Y_count)
allN <-sum(tagalog_stats$N_count)
totalWords <-allY+allN

prop.test(x=allY, n=totalWords, alternative = "greater")
```

```
##
##  1-sample proportions test with continuity correction
##
## data:  allY out of totalWords, null probability 0.5
## X-squared = 122.23, df = 1, p-value < 2.2e-16
## alternative hypothesis: true p is greater than 0.5
## 95 percent confidence interval:
##  0.6419931 1.0000000
```

```
## sample estimates:
##         p
## 0.6660682
```

Conclusion

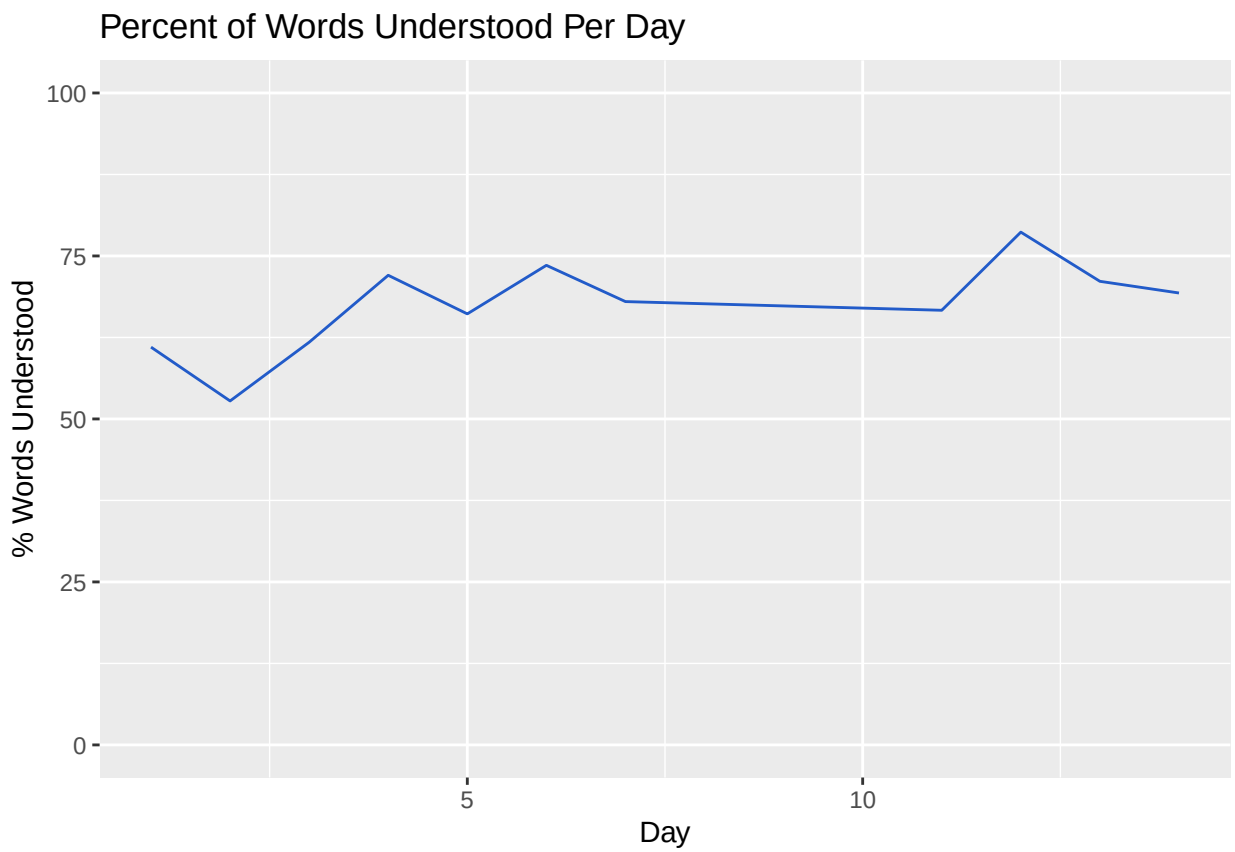
With a $p - value = 2.2e{-}16 < 0.05$, then with 5% significance level, the null hypothesis can be safely rejected; which supports the hypothesis that the number of words I understood is higher than the number of words I got wrong. This combined with a line graph shows a mild positive trend over time / an increase in words understood over time.

Plots

```
# Plot percentage of words understood per day as a line graph.
p <- ggplot(data = tagalog_stats, aes(x = Day, y = (percent_Y * 100)))+
  geom_line(color="#225BC9")+
  labs(title = 'Percent of Words Understood Per Day', x = 'Day', y = '% Words Understood')

p <- p + ylim(0,100)

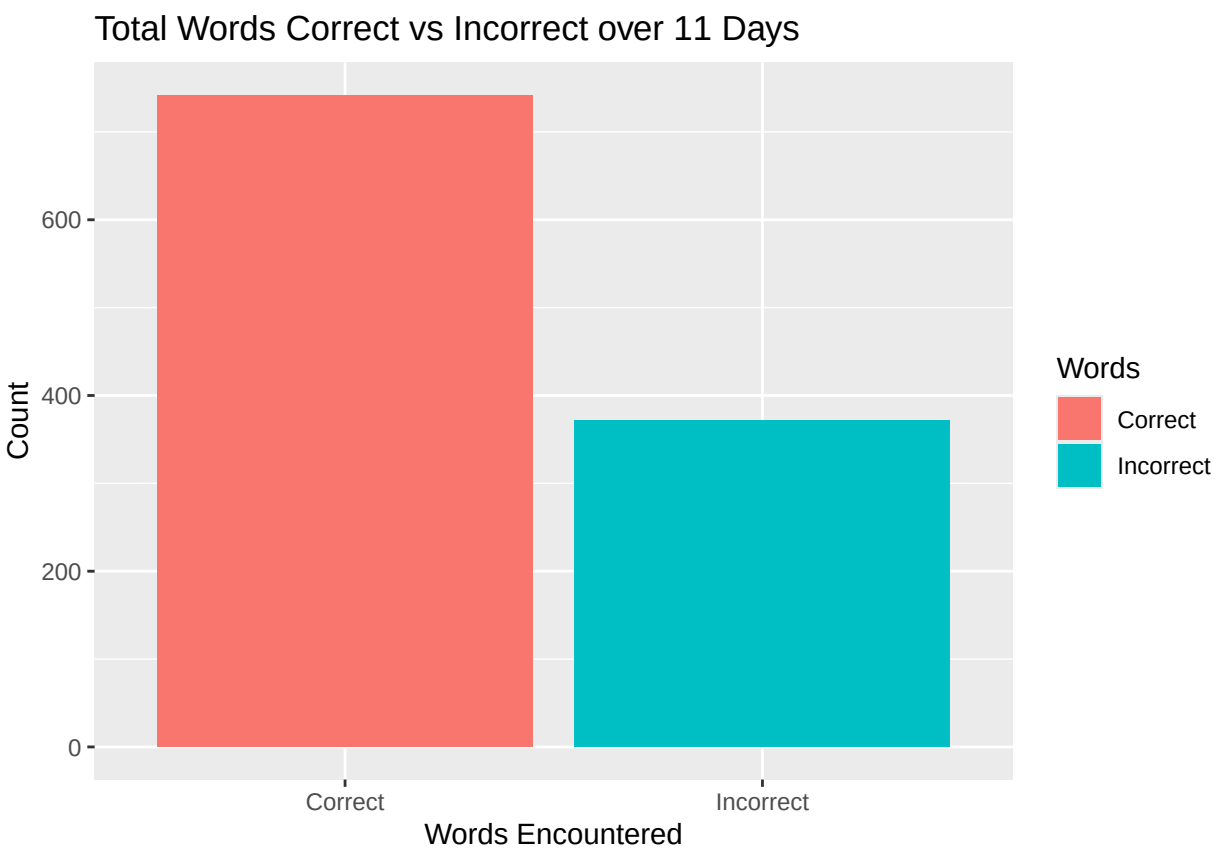
p
```



```r
# Plot total number of correct words vs incorrect words as a bar graph.
df <- data.frame(Words = c("Correct", "Incorrect"), Count =c(allY, allN))

p2 <- ggplot(df, aes(x=Words, y=Count))+geom_bar(stat="identity", aes(fill=Words))+labs(title="Total Wo

p2
```




```r
#plots show a small but clear positive trend
```